

Yashraj Kupekar

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EDUCATION

Scaler School of Technology

Bachelor of Computer Science

Bangalore, India

July 2027

EXPERIENCE

Data Science Intern

Motion Education

Remote

Jun 2025 – Oct 2025

- Built an end-to-end LLM-powered chatbot for the Motion Learning app serving real learners, using LangChain and LangGraph with ReAct-style agents, agentic routing, agentic-RAG, and tool orchestration.
- Designed multi-turn stateful conversation flows with OpenAI function calling, context management, and cost-aware prompting; shipped to production with async FastAPI, MongoDB, Pinecone, Docker, and Opik for observability.

Software Development Intern

EasyTradeWare

Remote

Dec 2024 – Feb 2025

- Built secure APIs and automation workflows for cryptocurrency trading, including Binance BTCUSDT data processing and order book operations.
- Contributed to a multi-threaded C++ trading system with WebSocket-based real-time data handling.

PROJECTS

Medicine Assistant – Offline Voice AI for Pharmacy Workflows | GitHub

- Built a real-time offline edge pipeline: Silero VAD → Whisper STT → local ReAct agent (Llama 3.2 via Ollama) → SQLite + FAISS/ChromaDB retrieval → Piper TTS – full task-oriented voice interaction with zero cloud dependency.
- Designed layered test suite (test_layer1–5) covering STT accuracy, agent tool calls, retrieval quality, and end-to-end pipeline; documented architecture in CONCEPTS.md and system-design diagram.

Guftagu – Multi-Character Voice Conversational System | GitHub

In Progress

- Building a 3-agent LangGraph pipeline (Writer → Dialogue Writer → Actor) for multi-character, multi-turn story-driven conversations with conditional routing, MongoDB-backed session persistence, and Sarvam AI for Indian-language TTS/STT (Hindi, Marathi, Hinglish).
- Researching and solving agent problems like non-coherent long conversations, character shifting, etc

Boardroom.ai – Multi-Agent Solopreneur Assistant | GitHub

- Designed and built the FastAPI + LangGraph backend: supervisor agent routing to specialist subgraphs (Finance, Market Research, Content) with iteration-capped looping, MongoDB thread checkpointing, and thread-safe /chat endpoint; Dockerized and deployed to production.

Retail Agent – Multi-Agent Retail Conversation Intelligence | GitHub

- Built a voice-to-insight pipeline: speaker-diarized transcription → Staff Performance Agent → Demand Analysis Agent → structured retail report; exposed via async FastAPI with thread-based session continuity and a tests/ suite.

Durable Webpage Builder Agent | GitHub

- Fault-tolerant AI agent using Temporal workflows – LLM planner routes to retryable specialist activities (HTML, CSS, JS, QA) with Signals/Queries for real-time user interaction; LiteLLM backend supports OpenAI, Gemini, and Anthropic.

SKILLS

Languages: Python, C++, Java, JavaScript

LLM APIs & Tooling: OpenAI API, Anthropic Claude API, function calling, context management, cost-aware prompting, LiteLLM

Agentic Frameworks: LangGraph, LangChain, ReAct agents, multi-agent supervisor patterns, tool orchestration, agentic-RAG

Voice / Multimodal: Whisper, Silero VAD, Piper TTS, Sarvam AI (STT/TTS), speaker diarization

RAG & Vector DBs: Pinecone, FAISS, ChromaDB – chunking, retrieval strategies, agentic retrieval

Infra & Serving: FastAPI, MongoDB, SQLite, Docker, Temporal, Opik (observability)

CERTIFICATIONS

Machine Learning Specialization – Coursera | Machine Learning A-Z – Udemy